

Project: How does the performance of a Greenpower car change during a race?

My team and I are going to use data from our performance at Dunsfold race on 13/09/2026 to analyse how the performance of Empress (our car) changes during a race.

Main objectives:

- See what conclusions can be made about the performance of empress.
- Identify the source of what causes empress to slow down during race.
- Use the data to make the process of looking for solutions that allow empress to maintain speed for as long as possible during races easier.

Display Table:

```
GreenpowerRaceData_real_ = readtable('Greenpower Dunsfold race data.xlsx');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

GreenpowerRaceData_real_

GreenpowerRaceData_real_ = 38×6 table

	LapNumber_1	LapTime_secs__1	LapNumber_2	LapTime_secs__2
1	1	119.9000	1	116.7000
2	2	465	2	111.3000
3	3	113.3000	3	109.5000
4	4	111.2000	4	111.5000
5	5	109.7000	5	108.9000
6	6	113.2000	6	113.9000
7	7	115.2000	7	113.9000
8	8	267.9000	8	109.3000
9	9	111.6000	9	109.6000
10	10	111.5000	10	111.9000
11	11	112.7000	11	113.7000
12	12	111.9000	12	112.4000
13	13	112.8000	13	283.5000
14	14	113.8000	14	121.3000

⋮

Import data from table into variables:

```
lapNumber1 = GreenpowerRaceData_real_.LapNumber_1
```

```
lapNumber1 = 38×1
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
⋮
```

```
lapTime1 = GreenpowerRaceData_real_.LapTime_secs__1
```

```
lapTime1 = 38×1
```

```
119.9000  
465.0000  
113.3000  
111.2000  
109.7000  
113.2000  
115.2000  
267.9000  
111.6000  
111.5000  
112.7000  
111.9000  
112.8000  
113.8000  
113.7000  
⋮
```

```
lapNumber2 = GreenpowerRaceData_real_.LapNumber_2
```

```
lapNumber2 = 38×1
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14
```

```
15  
:  
:
```

```
lapTime2 = GreenpowerRaceData_real_.LapTime_secs__2
```

```
lapTime2 = 38×1  
116.7000  
111.3000  
109.5000  
111.5000  
108.9000  
113.9000  
113.9000  
109.3000  
109.6000  
111.9000  
113.7000  
112.4000  
283.5000  
121.3000  
116.6000  
:  
:
```

```
lapDistance = GreenpowerRaceData_real_.TrackDistance_metres__
```

```
lapDistance = 38×1  
1210  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
:  
:
```

```
totalRaceTime = GreenpowerRaceData_real_.TotalRaceTime_mins__
```

```
totalRaceTime = 38×1  
90  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN
```

```
NaN
NaN
⋮
```

Calculate Average lap time:

```
avgLapTime1 = mean(lapTime1(1:34))
```

```
avgLapTime1 =
153.0588
```

```
avgLapTime2 = mean(lapTime2)
```

```
avgLapTime2 =
159.5000
```

Calculate Fastest lap:

```
fastestLap1 = min(lapTime1(1:34))
```

```
fastestLap1 =
109.7000
```

```
fastestLap2 = min(lapTime2)
```

```
fastestLap2 =
108.9000
```

Calculate Slowest lap:

```
slowestLap1 = max(lapTime1(1:34))
```

```
slowestLap1 =
465
```

```
slowestLap2 = max(lapTime2)
```

```
slowestLap2 =
807.2000
```

Calculate difference between fastest and slowest lap:

```
diff1 = slowestLap1 - fastestLap1
```

```
diff1 =
355.3000
```

```
diff2 = slowestLap2 - fastestLap2
```

```
diff2 =
698.3000
```

Calculate the speed of each lap:

```
lapDistanceDivider = 1210; % metres
```

```
eachLapSpeed1 = lapDistanceDivider./lapTime1
```

```
eachLapSpeed1 = 38×1  
10.0917  
2.6022  
10.6796  
10.8813  
11.0301  
10.6890  
10.5035  
4.5166  
10.8423  
10.8520  
10.7365  
10.8132  
10.7270  
10.6327  
10.6420  
:  
:
```

```
eachLapSpeed2 = lapDistanceDivider./lapTime2
```

```
eachLapSpeed2 = 38×1  
10.3685  
10.8715  
11.0502  
10.8520  
11.1111  
10.6234  
10.6234  
11.0704  
11.0401  
10.8132  
10.6420  
10.7651  
4.2681  
9.9753  
10.3774  
:  
:
```

Calculate Average lap speed:

```
avgLapSpeed1 = lapDistance(1)/avgLapTime1
```

```
avgLapSpeed1 =  
7.9055
```

```
avgLapSpeed2 = lapDistance(1)/avgLapTime2
```

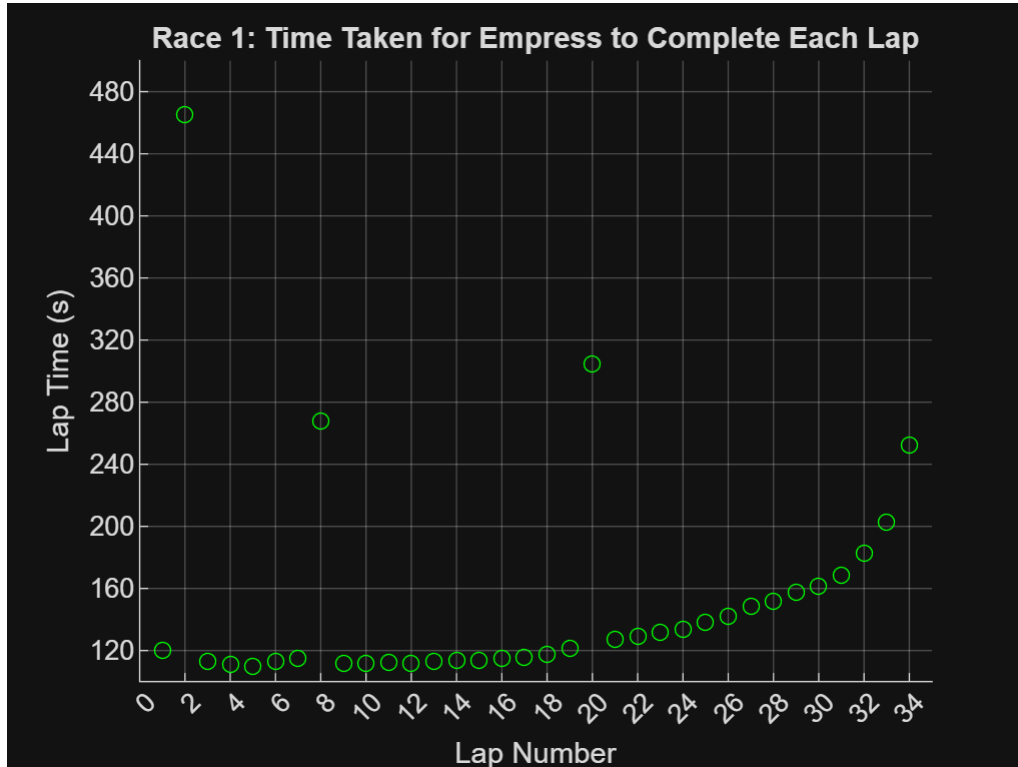
```
avgLapSpeed2 =  
7.5862
```

Graph to display the time taken for empress to complete each lap in Race 1:

```

scatter(lapNumber1, lapTime1, "g", "o")
ylabel("Lap Time (s)")
xlabel("Lap Number")
title("Race 1: Time Taken for Empress to Complete Each Lap")
set(gca, "xtick", 0:2:34)
set(gca, "ytick", 0:40:500)
grid on

```

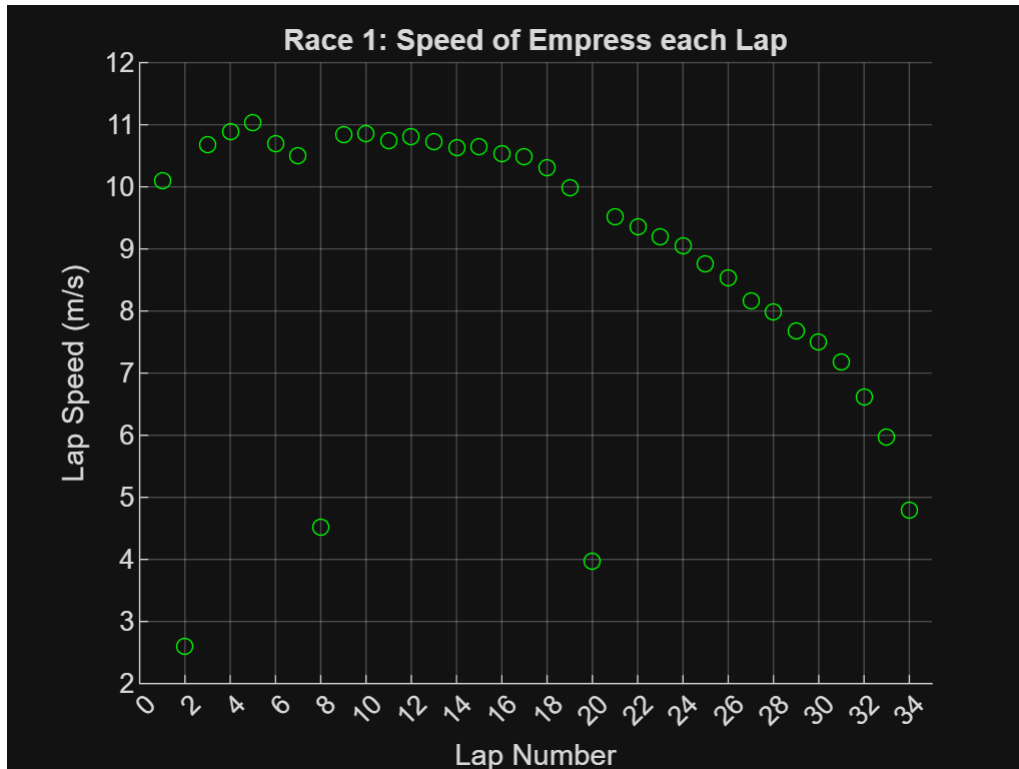


Graph to display the speed empress had each lap in Race 1:

```

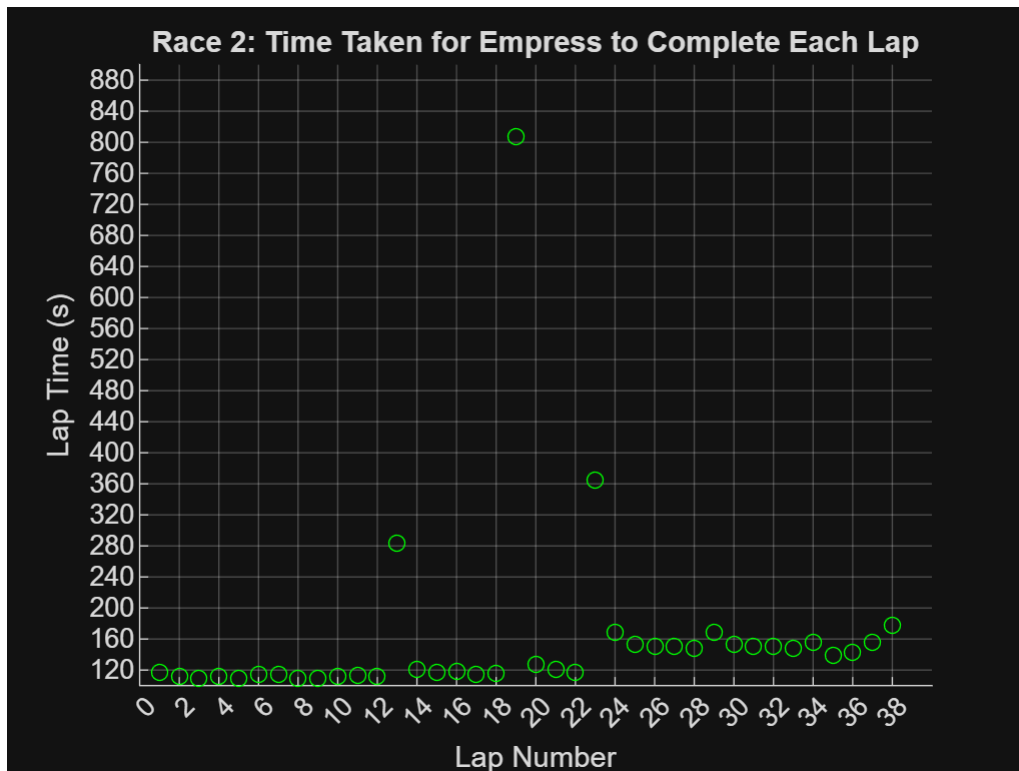
scatter(lapNumber1, eachLapSpeed1, "g", "o")
ylabel("Lap Speed (m/s)")
xlabel("Lap Number")
title("Race 1: Speed of Empress each Lap")
set(gca, "xtick", 0:2:34)
set(gca, "ytick", 0:1:12)
grid on

```



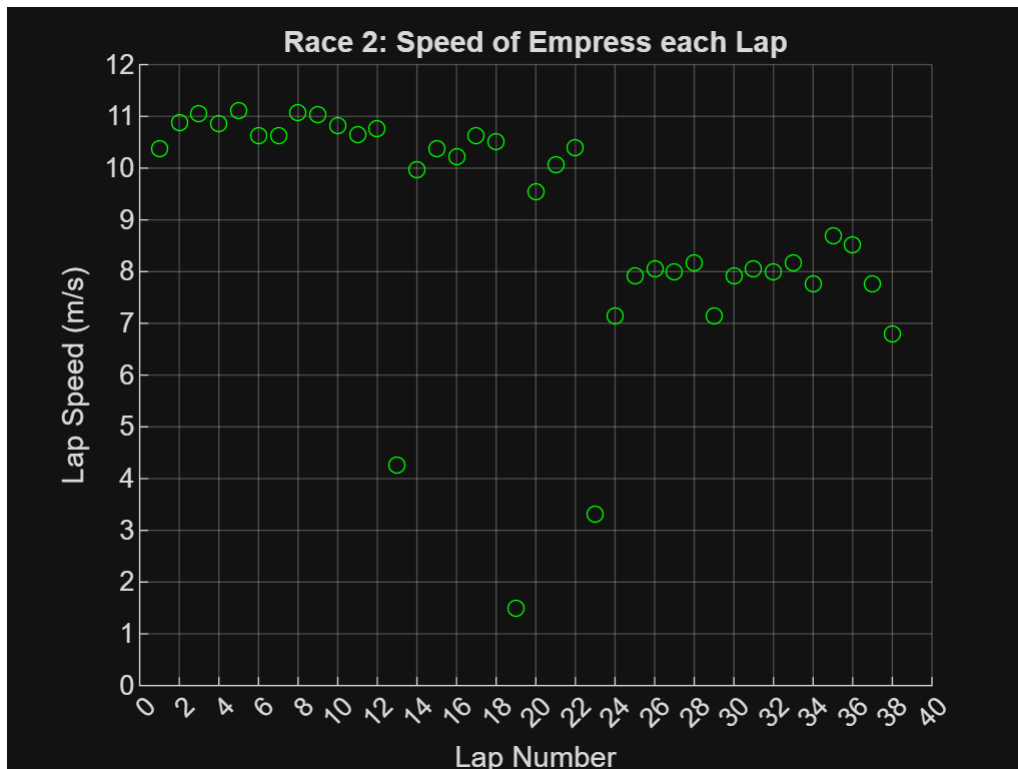
Graph to display the time taken for empress to complete each lap in Race 2:

```
scatter(lapNumber2, lapTime2, "g", "o")
ylabel("Lap Time (s)")
xlabel("Lap Number")
title("Race 2: Time Taken for Empress to Complete Each Lap")
set(gca, "xtick", 0:2:38)
set(gca, "ytick", 0:40:1000)
grid on
```



Graph to display the speed empress had each lap in Race 2:

```
scatter(lapNumber2, eachLapSpeed2, "g", "o")
ylabel("Lap Speed (m/s)")
xlabel("Lap Number")
title("Race 2: Speed of Empress each Lap")
set(gca, "xtick", 0:2:40)
set(gca, "ytick", 0:1:12)
grid on
```



Empress' place in each race:

Race 1: 16th

Race 2: 11th

Analysis of graphs:

Race 1:

Conclusions from Graph titled "Race 1: Time Taken for Empress to Complete Each Lap":

The time taken for the car to complete each lap is quite steady around 110seconds (despite some anomalies which are from pit changes) until lap 19 out of 34 where it starts increases at an increasing rate, until it reaches around 250seconds in lap 34. This is more double the initial time taken.

Therefore, Empress' performance declined. The time taken was quite consistent in one section, but in the second section it increased steadily.

Conclusions from Graph titled "Race 1: Speed of Empress each Lap":

The speed of the car is quite constant between 10m/s and 11m/s (despite some anomalies which are from pit changes) until lap 19 out of 34 where it starts to decrease at an increasing rate until it's speed is about 5m/s. This is half the initial speed.

One area of note that Sophie and I noticed is a dip in speed from lap 5 to 6 and 7 just before a pit change. After this pit change, the speed went back up to it's previous level. This could suggest it would have been better to do the pit change after lap 5 and allow the motor to cool, to remove this dip in speed. However it is difficult to

determine where this optimum pit change times could be in races overall. Using an eChook (data logging tool to log current, voltage etc.) during races could help determine this time.

Empress' performance declined. The speed was quite consistent in one section, but in the second section it decreased steadily.

Race 2:

Conclusions from Graph titled "Race 2: Time Taken for Empress to Complete Each Lap":

The time taken for the car to complete each lap is quite steady around 110 seconds (despite some anomalies from pit changes and one from a red flag, which stopped the race for a period of time) until lap 24 out of 38, where it jumps up to around 160 seconds and remains here until lap 38. This time taken is higher compared to the initial time taken.

Therefore, Empress' performance declined but had consistent time taken split into two different sections.

Conclusions from Graph titled "Race 2: Speed of Empress each Lap":

The speed of the car is quite constant between 9.5m/s and 11m/s until lap (despite some anomalies) until lap 24 out of 38 where it jumps up to around 7m/s to 8.5m/s and remains here until lap 38. This speed is lower compared to the initial speed.

Therefore, Empress' performance declined but had consistent speed split into two different sections.

Overall analysis:

Which race did Empress have a better performance in and why?

Empress performed better in Race 2.

In race 1, the speed was quite consistent in one section, but in the second section it decreased steadily. In race 2, had consistent speed split into two different sections, where the second section was around a lower speed than the first.

Since Empress was able to maintain more consistency in Race 2, it performed better in race 2.

Furthermore, in race 1 empess started at 10m/s and finished at 5m/s, but in race 2 empess started at around 10m/s and ended at 8.5m/s. Therefore, Empress performed better in race 2.

One reason why she could have performed well is because the batteries used in the second race had higher Amp-hours. So, more charge was available for the car to use in the second race.

Another reason why she could have performed well is because we realigned the motor sprocket with the axle sprocket between race 1 and 2. However, this does not provide a clear reason why the speed was more consistent in the second race.

Also, it rained in the second half of race 2, which caused Empress to move slower because of added resistance. The drivers' apprehension on holding down the power button in fear of crashing the car could also be a factor, which slowed down the car. Despite this, Empress is faster in race 2.

What was the cause of reduction of speed in both races?

Saffy, car lead of KM1 (the fastest car our school has), suggested that the decrease in speed in both races is due to the aerodynamics and weight of our car. Our car is not completely streamlined and bullet shaped, which may have caused the speed to be low. Our car's chassis is also quite heavy compared to KM1 because when it was built it was filled with too much resin which made it too heavy. Poor aerodynamics and heavy weight are two factors which would definitely cause the car to be slower.

A possible solution to this is trying to make our car more aerodynamic by using less structural foam or shaving it down to make the car more sleek.

We can streamline the vehicle's silhouette by tucking the protruding wheels into the main body.

In regards to the weight of the car we cant change the chassis but we can try and identify and remove unnecessary mass from the car.

Extra tasks to look into:

Mr Bamford spoke about how the fastest teams' cars are able alter their motor to their benefit.

In the car, drivers can turn a dial to reduce their battery charge by one amp when they see a pit sign with this direction during a race. This will reduce their speed but conserve battery for later.

Just before the penultimate lap another pit sign will be shown to direct them to increase their car by one amp again, which will give them much more speed in this lap.

This is desirable because Greenpower count the last lap in the following way. Using an example to explain; say the penultimate lap is 4minutes, the last lap is calculated to be 1/2 of this lap. If the penultimate lap is 2 minutes the last lap is calculated to be 1/4 of this lap.

Therefore, it is better to have a a shorter amount of time in the penultimate lap than other laps previously in the race, so you can have a short time in the last lap of the race. Perhaps we could add a system to do this to Empress.

Next steps:

After consulting with the team, here are some next steps we would like to move forward with.

To help with aerodynamics, a smooth covering made of lightweight vinyl over empress should help to improve aerodynamics. If our school goes to the next race, Goodwood we should prioritise this.

At the moment, Empress' wheels are very slightly "toe in". If we get to the finals, we need will shorten the track rods on the front steering to get them perfect.

Export code and results as a pdf

```
export("Dunsfold_2026_empress.mlx")
```

```
ans =  
'/MATLAB Drive/Dunsfold_2026_empress.pdf'
```

